MEMO October 2026

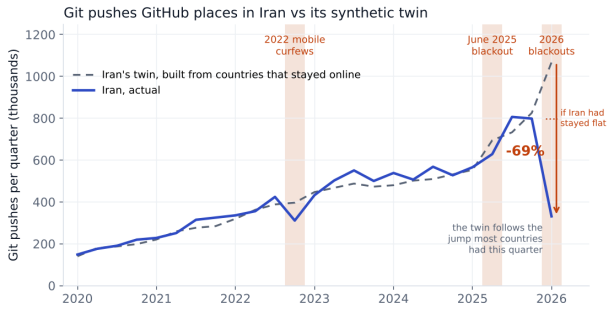
What happens to a country's GitHub activity when its internet goes dark?

From Matin Nikoomanzar. GitHub Innovation Graph, 106 comparison countries, 2020 to early 2026. Full paper, code and tests at github.com/MatinNikoo/internet-shutdowns

Bottom line. Iran was offline for about half of early 2026, and the git pushes GitHub places in Iran came in 69% below a synthetic twin, or 59% below if Iran would have stayed flat while most of the world jumped that quarter. Myanmar after its 2021 coup is the only other drop that big. This is GitHub's map and says nothing about the economy. Some of the missing activity is people on VPNs being counted in other countries. Shutdowns of a week or so can't be seen in quarterly data.

What I found

- **Two big drops.** Iran 2026 (−69%, 95% interval −76% to −61%) passes every check I fixed before running them. Myanmar 2021 (−56%) passes every one that can be run, but it has four quarters of history and those checks are weak for it. Sudan 2021 (−32%) fails backdating.
- **Dates from data that isn't GitHub.** Daily OONI network tests put Iran's 2026 blackout at 49 days, inside the 48 to 58 in my registry. Wikipedia pageviews date Bangladesh 2024 and Tanzania 2025 to the day.
- **Some activity moved.** Repository counts almost never fall (more than 2% in 0.9% of country quarters). Iran's changed by −4.5% and Myanmar's by −14.2%.
- **Crises move it too.** Israel after 7 October 2023, with no shutdown, came in at −23%. A drop of 20% can't be pinned on the internet.
- **Short shutdowns.** The four full shutdowns of 11 days or less pool to −7.8% (−19% to +5%), so I can't see them.



The twin is a weighted mix of countries that moved like Iran before the event. The gap after is the estimate.

What I got wrong before

- The first version said "each day offline cost 1.3 days of work". 89% of that slope is one event, and it falls to 1.02 if Iran missed the 2026 jump. With OONI's count of disrupted days the slope is 1.39 (1.10 to 1.67). It is positive, but the interval is wider than the 0.5 my plan required before saying anything about a slope of 1, and without Iran 2026 it is 0.53. So I don't say anything about it.
- It said developers "fell" 6% in Iran. That was a gap to the twin. The count still grew.

How sure I am

- Three estimators, four alternative donor pools and leave one out agree within a few points on both big drops. Iran 2026 survives a break from the twin's path 3.4 times the worst one in Iran's own history. Without the 24 donors that Access Now lists with a full shutdown (Ethiopia is one), Iran 2026 is −69% and Myanmar −55%.
- 0 of 10 crises without a shutdown came out significant. Held out pre quarters stay inside the placebo band for Iran 2026. I chose those ten crises after writing the plan and before running them.
- A VPN test in likely exit countries has no power. It could miss the whole amount 23 times over.

What I'd do with this. Study shutdowns with daily data, because quarterly country totals blur them. Treat a drop in attributed activity as partly a change in measurement. And don't compare a shutdown quarter with the quarter before when the whole world is moving.